PRATHAMESH KULKARNI

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ANALOG LAYOUT DESIGN ENGINEER

OBJECTIVE

Seeking a challenging role as an IC Layout Design Engineer and utilize my experience to deliver a project from concept to silicon.

PROFILE

- **Mask Design Engineer ● 2022 July – Till Date ● INTEL Corporation**
- **Component Design Engineer ● 2020 February – 2022 July ● INTEL Corporation**
- **Layout Engineer (Contract) ● 2018 April – 2020 February ● Western Digital Corporation through JGD Technology Pvt Ltd**
- **Analog layout training: ChipEdge Technology Pvt Ltd ● 2017 Sept – 2018 Jan**
- **Trainee at Infosys Mysore ● 2017 Feb – 2017 Aug**
- **Internship program in Schneider Electric Pvt Ltd ● 2016 July – 2016 Sept**

5 plus years of experience in custom Analog/mixed signal layout and test chip development. A good team player, fast learner and believes that meticulous effort upheld by sheer commitment and hard work is the key to success.

PROFESSIONAL EXPERIENCE

INTEL Corporation

Technology: N5 TSMC
Blocks: Clock critical blocks (clk distortion correction blocks) where the delay was critical and must meet the post layout simulation results.
Challenges: To meet the Post layout simulation results, connections are improved
Made Robust connection and met the post layout simulations in one go

Technology: N3 TSMC
Blocks: Completely handled test-row layout design using pad row, leaf cell layout design (array of PMOS and NMOS)
Challenges: Floor planning, area optimization, critical DRC fixes, signal planning, power planning, metal stack

Technology: N5P TSMC
Blocks: Completely handled test-row layout design using pad row, leaf cell layout design (array of PMOS and NMOS),
Challenges: Floor planning, area optimization, critical DRC fixes, signal planning, power planning, metal stack

Technology: N5 TSMC
Blocks: Completely handled test-row layout design using pad row, leaf cell layout design (array of PMOS and NMOS) Documentation
Challenges: Floor planning, area optimization, critical DRC fixes, signal planning, power planning, metal stack

Technology: N5 TSMC
Blocks: Completely handled test-row layout design using pad row, leaf cell layout design (array of PMOS and NMOS), Documentation, implementation of 3 diff flavors (lvt, svt, ulvt)
Challenges: Floor planning, area optimization, critical DRC fixes, signal planning, power planning, metal stack

Technology: C16ffc Samsung
Blocks: Completely handled test-row layout design using pad row, leaf cell layout design (array of PMOS and NMOS), Documentation
Challenges: Floor planning, area optimization, critical DRC fixes, signal planning, power planning, metal stack upto M11 for power routing

Technology: N6 TSMC
Taskforce: Completed extraction for 20 test rows consisting of 9 ring oscillators per row in a limited time
Challenges: time constraint

JGD Tech Pvt Ltd
Client: Western Digital Corporation, April 2018 – Feb 2020

1. **Assignments**: Bias generator, voltage reference generator, Op-amp for temperature sensor, CUA/CNA Standard cells including macros, VCC monitor (Op amp, switches), digital blocks, and resistor trimmer circuit.

   Training on: Floor planning, matching devices, power/signal routing, top level integration, shielding, density fixes, verifications (LVS & DRC)

2. **Project**: BICS5 X4 Memory, CNA (April 2018 - Present)
   Technology used: BICS5 NAND FLASH with 140nm nmos technology
   - **Digital Blocks**: Standard cells (AND, NOR, NAND, BUFFER, XOR, MUX) with different drive-strengths
     - **Challenges**: Area reduction, PIN accessibility, single metal routing.
   - **Analog Blocks:** Completely handled blocks like voltage regulators and corresponding digital logic blocks, high precision RC oscillators, low voltage charge-pump circuit, IO pads, top level signal routing
     - **Challenges:** Critical shielding and matching of clock signals, area constraint, routing track challenges dew to very few metal layers, consideration of EM effect in layouts of high-power blocks, symmetry of layout in high-speed clock signals, top level integration routing for IO- pads and analog blocks. Density fixes as per metal constraints
   - **Status of Project status:** Verification for the chip is completed, processed for tape out
3. **Project:** BICS6 Memory, CNA (2018-19)

Blocks: Relay Voltage Monitor
Challenges: PIN accessibility, signal routing, Verification.
Status of Project: Tape-out successfully completed

TECHNICAL SKILLS

TOOLS

- ***Layout Editor Tools***: Virtuoso Layout Editor (Cadence), IC compiler (Synopsys)
- ***Verification Tools:*** One LV(Intel), RC extraction, Caliber (Mentor Graphics), IC Validator (Synopsys)
- Exposure to various technologies including TSMC 3nm, n5 & 5P nm, 16nm, TMC->BiCS6 NAND FLASH with 140 nm,
- Unix/Linux/Microsoft operating systems

RECOGNITION:

- Significant Effort or Achievement Award in N5 Tapeout
- Notable Effort or Achievement Award for test chip extraction and target simulation support
- Notable Effort or Achievement Award for quality and timely delivery of test-chip
- Significant Effort or Achievement Award in N3 Tapeout

TRAINING

- **Institute:** ChipEdge Technologies Pvt. Ltd
- **Duration:** 5 months
- **Technology**: 90nm
- **Tools**: Synopsys IC-Complier and IC-validator
- **Projects**: Digital standard cell library, Low dropout Regulator (LDO), single stage OPAMP's, Current Mirrors and Multipliers.

EDUCATION PROFILE

- B. Tech in Electronics & Instrumentation Engineering, VTU -7.01 CGPA (2012 to 2016)
- Pre-University Education, Karnataka- 60.16% (2010 to 2012)
- X class from State Board, Karnataka- 76.32 % (2010)

PERSONAL PROFILE

Father's Name	**:** Late Vijeyendra Kulkarni
Date of Birth	**:** 12th September 1994
Marital Status	**: M**arried
Nationality	**:** Indian
Languages Known	: English, Kannada, and Hindi